Jordi Cerdà-Bautista

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Research Interests

Causal machine learning applied to food security, climate variability, conflict, and humanitarian policy in low- and middle-income countries. Methodological focus on causal effect estimation, causal discovery, and spatial/panel methods. Substantive focus on sub-Saharan Africa, with operational links to WFP, JRC, and FAO.

Education

Image Processing Laboratory (IPL), Universitat de València — **València, Spain**
Ph.D. in Remote Sensing — *Sep. 2022–Present*
Thesis: *Causal Machine Learning for Food Security: Evidence from Socio-Environmental Shocks*
Supervisors: Gustau Camps-Valls & Vasileios Sitokonstantinou
Defense expected: early 2027

Universitat de València — **València, Spain**
M.Sc. in Data Science — *2020–2022*
Thesis: *Deep Learning Methods for Atmospheric Correction of Satellite Images*
Supervisors: Luis Gómez Chova & Gonzalo Mateo García

Università degli Studi di Torino — **Turin, Italy**
Erasmus Programme — *2018–2019*

Universitat de València — **València, Spain**
B.Sc. in Physics — *2015–2020*
Thesis: *Interference with a spatially incoherent polarized source*
Supervisors: Genaro Saavedra Tortosa & Juan Carlos Barreiro Hervás

Honors & Awards

WCRP Open Science Conference — **Kigali, Rwanda**
Best In-Person Poster Award — *Oct. 2023*
Poster: *Understanding food insecurity in Africa through data-driven causal inference methods.*

Experience

Image Processing Laboratory (IPL), Universitat de València — **València, Spain**
Doctoral Researcher — *Feb. 2022–Present*
Causal machine learning for food security across sub-Saharan Africa: humanitarian intervention evaluation, climate–price interaction analysis, conflict spillover effects, and micronutrient pathway decomposition. Co-developer of *CauseMe*, a web platform for causal discovery in Earth and climate sciences.

Capgemini S.A. — **València, Spain**
Data Scientist Intern — *Oct. 2021–Feb. 2022*
ML models, SQL, web design.

Publications

Peer-Reviewed Journal Articles

2024: J. M. Tárraga, E. Sevillano-Marco, J. Muñoz-Marí, M. Piles, V. Sitokonstantinou, M. Ronco, M. T. Miranda, **J. Cerdà**, G. Camps-Valls. *Causal discovery reveals complex patterns of drought-induced displacement.* **iScience** 27(9), 2024. *[Cell Press; JCR Q1]*

Manuscripts in Preparation

2026: **J. Cerdà-Bautista**, V. Sitokonstantinou, J.M. Veiga López-Peña, D. Piovani, J.M. Tárraga, G. Camps-Valls. *Credit Access is Associated with Improved Food Security in the Horn of Africa.* Manuscript under revision.

2026: **J. Cerdà-Bautista**, V. Sitokonstantinou, H. Durand, G. Varando, M. Ronco, G. Camps-Valls. *Climate Sensitivity Modulates the Food-Security Impact of Price Spikes.* Manuscript under revision.

Preprints

2024: V. Sitokonstantinou, E. D. S. Porras, **J. Cerdà-Bautista**, M. Piles, I. Athanasiadis, G. Camps-Valls. *Causal Machine Learning for Sustainable Agroecosystems.* arXiv:2408.13155.

Peer-Reviewed Conference Proceedings

2024: **J. Cerdà-Bautista**, J. M. Tárraga, V. Sitokonstantinou, G. Camps-Valls. *Assessing the Causal Impact of Humanitarian Aid on Food Security.* IGARSS 2024 — IEEE International Geoscience and Remote Sensing Symposium, Athens.

Software

CauseMe: Web platform for causal discovery in Earth and climate sciences. Implements PCMCI, LinearVAR, Granger2D, and GraphEM with interactive graph and edge-bundling visualizations, and forecasting. causeme.uv.es

Causal4FS: Open-source pipelines for causal effect estimation on humanitarian and food-security data — covering cash-based transfers, price spikes, conflict escalations, and micronutrient pathways. github.com/jordicbau

Research Projects

2024–Present: *ThinkingEarth* — EU Horizon Europe Research & Innovation Programme. Copernicus Foundation Models for a Thinking Earth.

2023–2024: *Causal4Africa* — Microsoft Climate Research Initiative. Causal evaluation of humanitarian interventions in Africa.

2022–2023: *SCALE: Causal Inference in the Human–Biosphere Coupled System* — Fundación BBVA.

2022: *DeepCube* — EU Horizon 2020 (contributor). Explainable AI for Copernicus Earth Observation data.

Selected Presentations

Dec. 2025: Poster: *Causal Effects of Price Spikes on Food Insecurity*. EurIPS Workshop on Causality for Impact, Copenhagen.

Jun. 2025: Talk: *Human and Environmental Causal Effects on Food Security in Africa*. ESA Living Planet Symposium, Vienna.

Jul. 2024: Talk: *Assessing the Causal Impact of Humanitarian Aid on Food Security*. IGARSS, Athens.

May 2024: Poster: *Evaluating the Causal Impact of Humanitarian Interventions on Food Insecurity in Climate-Vulnerable Regions of Africa*. ESA EO4AGRI Workshop, Frascati.

Apr. 2024: Talk: *Causal Evaluation of Humanitarian Aid on Food Security*. EGU General Assembly, Vienna.

Oct. 2023: Poster (best in-person award): *Understanding Food Insecurity in Africa Through Data-Driven Causal Inference Methods*. WCRP Open Science Conference, Kigali.

Apr. 2023: Talk: *Causal Inference to Study Food Insecurity in Africa*. EGU General Assembly, Vienna.

Additional schools and workshops attended: Nordic Probabilistic AI School (NTNU, Trondheim, 2023); *When Causal Inference meets Statistical Analysis* (CNAM, Paris, 2023).

Technical Skills

Methods: Causal effect estimation; causal discovery; panel and spatial inference; refutation and sensitivity testing.

Programming: Python (primary), R, SQL, MATLAB.

ML & Causal Libraries: EconML, DoWhy, Tigramite, DoubleML, scikit-learn, TensorFlow, PyTorch, XGBoost.

Geospatial & EO: Google Earth Engine, xarray, rasterio, GeoPandas.

Engineering & Web: Git, PySpark, R Shiny, Dash, Flask, HTML/CSS/JavaScript.

Other: LaTeX, Microsoft Office.

Selected Training

Universitat de València: Causal Discovery and Inference; Kernel Methods in Machine Learning; Deep Learning for Earth Sciences.

Coursera & Pluralsight: Machine Learning with TensorFlow on Google Cloud (specialisation); Advanced Machine Learning on Google Cloud (specialisation); Apache Spark fundamentals; Google Cloud core infrastructure.

Languages

Catalan: Native

Spanish: Native

English: Fluent

Italian: Intermediate

References

Available on request.